Personalised Offer Engine: Progress Report

Navpreet Kaur · Hackathon 2, Punjab Jobs AI Bootcamp (J-PAL South Asia) · Report date: Thu 22 Oct 2026 · Version v0.1 · Submission due **Tue 27 Oct 2026, 12:00 noon**

Repo: github.com/Navkaur-codes/Offer-Engine · Bot: @navpreet_offers_bot (t.me/navpreet_offers_bot) · Live API: offer-engine-n4jx.onrender.com/health

Live	7	19	13	21	~12%
bot + backend deployed on Render	small commits on GitHub	automated tests passing	database tables designed	module interfaces defined	of total build (foundation done)

1. The project in brief

An **analytics and AI decisioning platform** for an e-commerce marketing team. It decides **which offer each customer should get, and when**, and delivers it through a Telegram bot. There is no store; customer data is fully synthetic. Marketers set the limits as plain-English rules. An AI agent (Groq Llama) chooses and personalises an offer within those limits, and **every AI choice is re-checked by rule code** before it is sent. If the AI fails, a deterministic rules fallback decides instead.

- **Core idea, "don't pay people who'd buy anyway":** uplift scores spend discount only where it changes behaviour, and a 10% holdout group proves the extra revenue.
- **Dynamic rules per customer type:** Global → Festival → Persona → Individual. Lower layers can only tighten.
- **Festival-aware offers** (Karwa Chauth, Dhanteras, Diwali, Gurpurab, Lohri) with cultural safety rules.
- **Explainable:** every decision is logged with why this customer, why this offer and which rules passed. Users can ask the bot /why.

2. Progress till now

Planning and design

- Brief analysed; full plan, architecture and feature priorities (must / should / wow) written.
- Agent design: guardrail harness, read-only tools, system prompt, retry, then rules fallback.
- Distinctive dashboard design system ("Ledger"), deliberately not a generic blue-green AI look.
- Work split into 4 parallel workstreams with fixed boundaries and shared data contracts.
- Project control docs: rules file (CLAUDE.md), progress log, session tracker, "detective" checklist of 28 common traps, ready prompts per phase.

Built, tested and deployed

- Shared data contracts (customer profile, offer, rule, decision, explanation, festival).
- Settings, India-time clock with demo time-travel, database layer (SQLite local / Postgres cloud).
- Plug-in interfaces with realistic placeholder data, so all 4 parts can be built in parallel.
- Backend with health check and a secured Telegram webhook (rejects wrong secrets).
- Bot live: /start replies. Hosted free (Render + Neon), kept awake by a 10-min ping.
- Security fixes: no secrets in git; bot token kept out of logs.

3. What comes next

Day	Data & scoring	Rules + AI agent	Telegram bot	Analytics dashboard
Fri 23	Customer personas, 12-month behaviour simulation, festival calendar	Layered rules engine + tests that no offer exceeds a limit	Onboarding, offer cards, opt-out	Design system, layout, kill switch
Sat 24	Customer scores: RFM, churn risk, policy	Guardrail harness + rules fallback	/simulate cart, browse, inactive… and /why	Overview + decision log
Sun 25	Uplift model, realism + experiment stats	Live AI agent + message checks	Customer Lab: 10 persona bots in a group	Rules editor with impact preview
Mon 26	Tuning to real-world benchmarks	Daily AI summary of results	Admin tools, polish	Experiments, festivals, playground

Mon 26, 18:00: feature freeze, then a test round with classmates, the README and screenshots. **Tue 27 morning:** final checks and submission.

4. What I will submit (Tue 27 Oct, before 12:00)

- **GitHub repository** with all source code and small, regular commits.
- **README**: setup, usage, a 5-minute demo script, and how each judging criterion is met.
- **CLAUDE.md** plus the docs folder showing how Claude Code was used.
- **Telegram bot username:** @navpreet_offers_bot, kept running until judging ends.
- **Live dashboard link** (Vercel) and backend health link.
- Final progress report and screenshots.

5. Risks and how they're handled

AI free-tier limits → rules fallback always sends, plus a daily AI budget guard. Free hosting sleeps → 10-minute ping. Festival dates → verified against a calendar. Spamming classmates → frequency caps and quiet hours (21:00–09:00).